Josh Mackenzie

Data Engineer · UK-based

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Data engineer with 5 years' experience building data platforms end to end, most recently as the sole data engineer at a carbon reporting company.

Skilled in Python, dbt, Azure and LLM-powered data tools, with a first-class degree in Electronic Engineering and a background in machine learning. Open to remote roles owning data platforms and shaping product.

EXPERIENCE

Data Engineer, NeuerEnergy (part of Aqovia) — Oct 2024 – Present

Sole data engineer in a small product team, with full ownership of the data platform and a lead-level role in product decisions.

- **Data engine:** Designed and built the core data platform from scratch, ingesting multi-source data, transforming it with dbt and DuckDB, and feeding the customer-facing product.
- **LLM bill extraction:** Built a multi-model OCR tool (Claude, Gemini, GPT) to extract data from energy bill PDFs, reaching 95%+ accuracy across 1,200 bills and saving hundreds of hours of manual work.
- **UN IOM:** Built the Python pipeline behind an LLM tool that classifies documents at scale for the International Organization for Migration.
- **Databricks:** Built the Databricks integration for enterprise clients, including a major international fashion brand.
- **Data management:** Owned data quality and governance, building dbt tests, schema checks and freshness monitoring with Dagster and Slack alerting, designing the client onboarding process, and managing access controls for compliance.
- **Infrastructure:** Managed Azure infrastructure as code with Terraform/OpenTofu, with Docker deployments through CI/CD in Azure DevOps.
- **Product & hiring:** Helped shape the roadmap, prioritising by ROI and aligning stakeholders on expectations and timelines; ran technical interviews for data engineering roles, with input on hiring decisions.

Tech
Python · SQL · dbt · DuckDB · Dagster · PostgreSQL · Databricks · Docker · Azure · Azure DevOps · Terraform/OpenTofu · LLM APIs

Data Engineer, InCrowd — Oct 2021 – Mar 2024

- Built and maintained hundreds of SQL and dbt pipelines on AWS for major sports teams and leagues, and automated workflows in Python and Prefect.
- Designed data models with client teams around their reporting needs.
- Designed and built an automated GDPR data deletion service, streamlining a complex legal compliance process.
- Set up pipeline monitoring and alerting, including data freshness checks.

Tech
Python · SQL · dbt · Prefect · PostgreSQL · AWS · Snowplow · Google Analytics · REST APIs

EDUCATION

BEng (Hons) Electronic Engineering with Nanotechnology, University of York — Sep 2016 – Jul 2021
First-Class Honours

- Final-year project: trained a tensegrity robot to move using soft actor-critic reinforcement learning in TensorFlow.
- Studied control systems, signal processing, semiconductors, photonics and chip design.
- Used Python, Java and C extensively for data analysis and hardware control.
- Led a team designing and building a multimeter.
- Fabricated and analysed nanoscale devices in a cleanroom.

VOLUNTEERING

Organiser, Limmud — Ran 3–7 day residential events for 200–3,000 attendees, covering catering, registration, accommodation, transport, tech and participant care. 2012 – 2024

Trustee, University of York Students' Union — Oversaw regulatory compliance and financial stability for a c.£5m organisation; introduced the Real Living Wage for staff. 2019 – 2021

President, University of York Electronics Society — Led the committee, ran weekly lab sessions, and organised events and mentoring for c.500 members. 2018 – 2021